

NoSQL

Databases

T2 – Oracle Overview

Teacher: José Machado / Cristiana Neto
Email: jmac@di.uminho.pt /
cristiana.neto@algoritmi.uminho.pt



Summary

- 1 Introduction to Oracle
- 2 SQL in Oracle
- 3 Oracle vs MySQL
- 3 Oracle NoSQL Database

Oracle Enterprise 19c

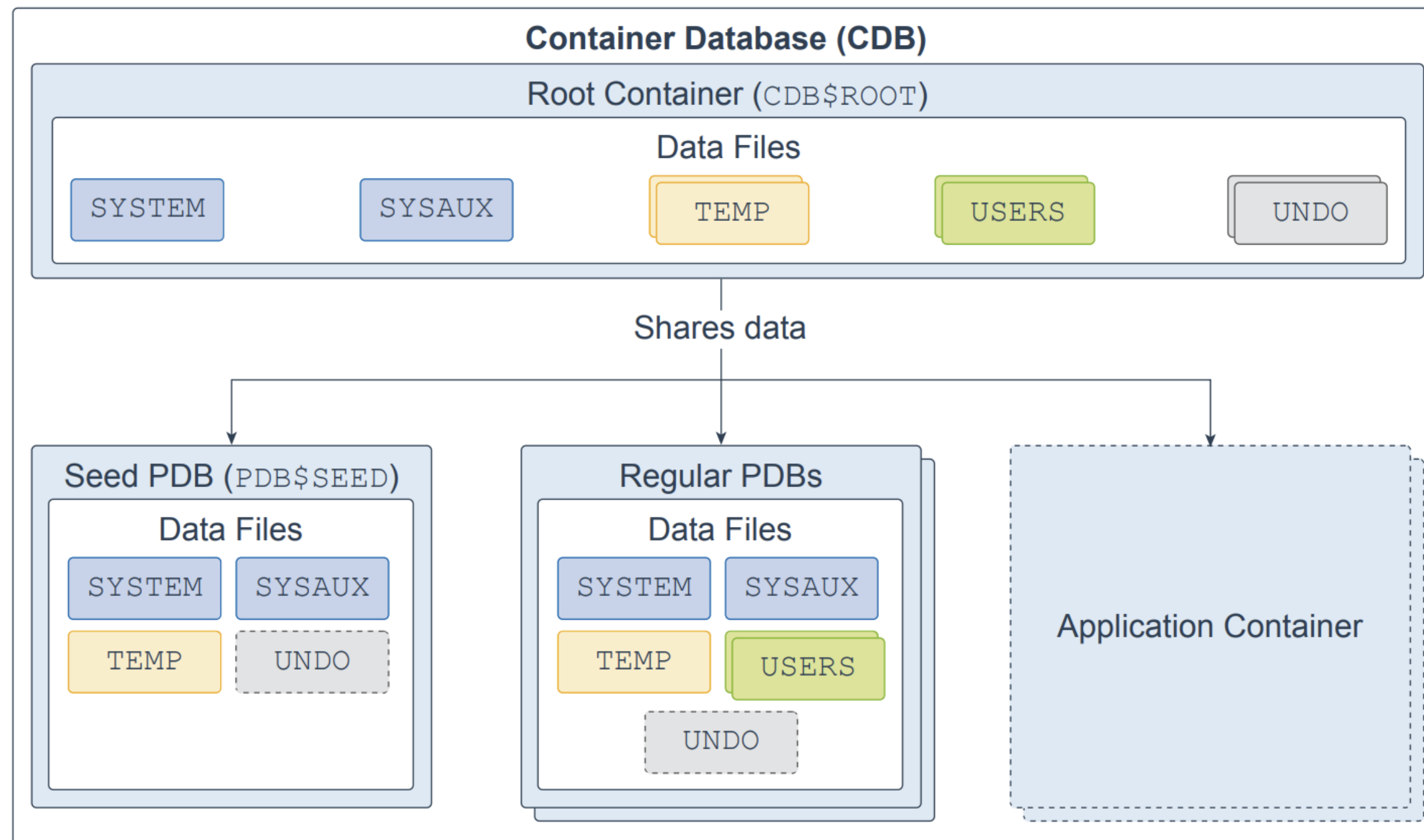
Oracle Database is a relational database management system (RDBMS) developed by Oracle Corporation. It is one of the most popular and widely used database systems in the world, known for its scalability, reliability, and robustness.



Oracle Database 19c is the current long-term release and provides the highest level of release stability and the longest period of time for support and bug fixes.

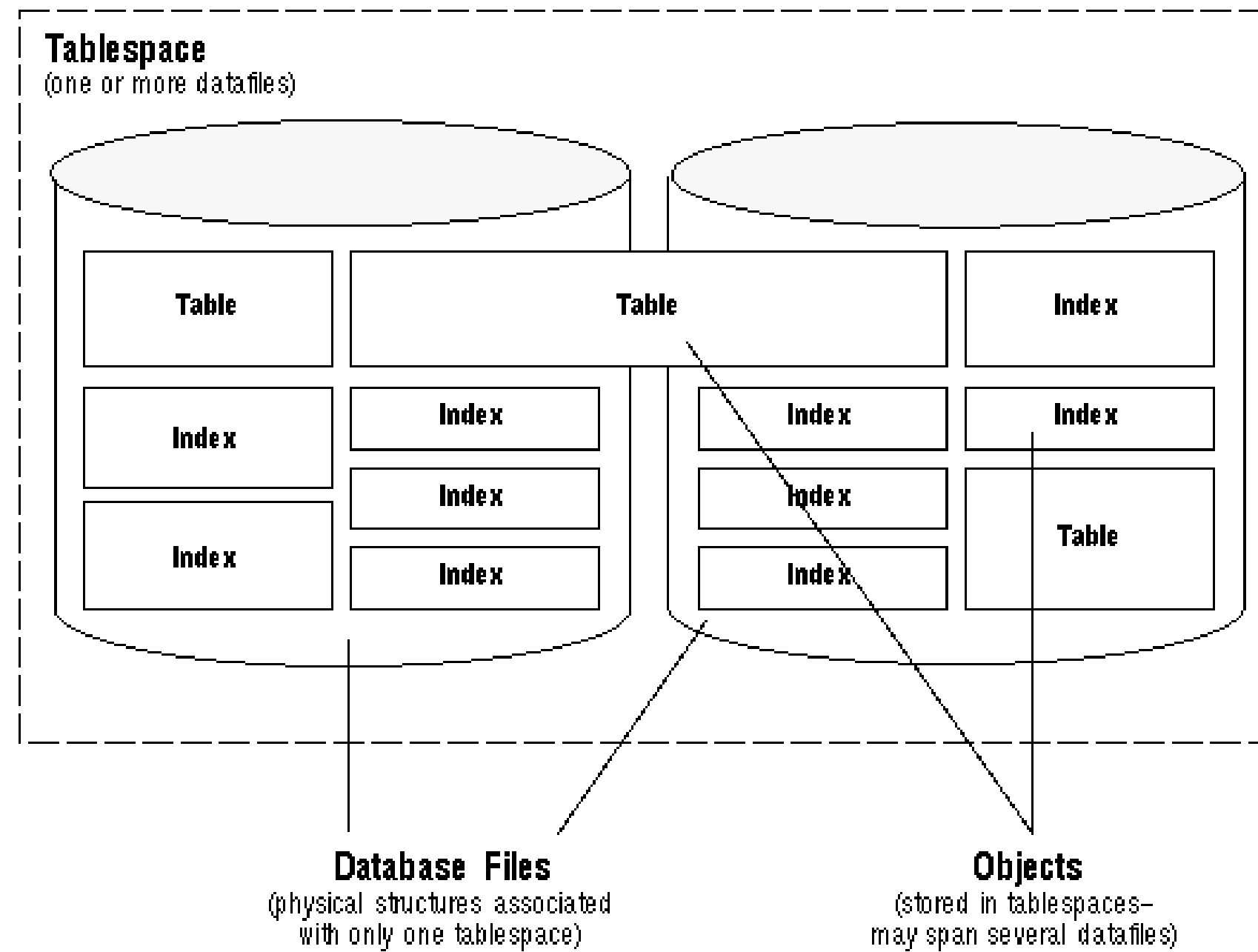
Oracle Enterprise 19c

➔ Architecture



Oracle Enterprise 19c

➔ Architecture



Oracle Enterprise 19c

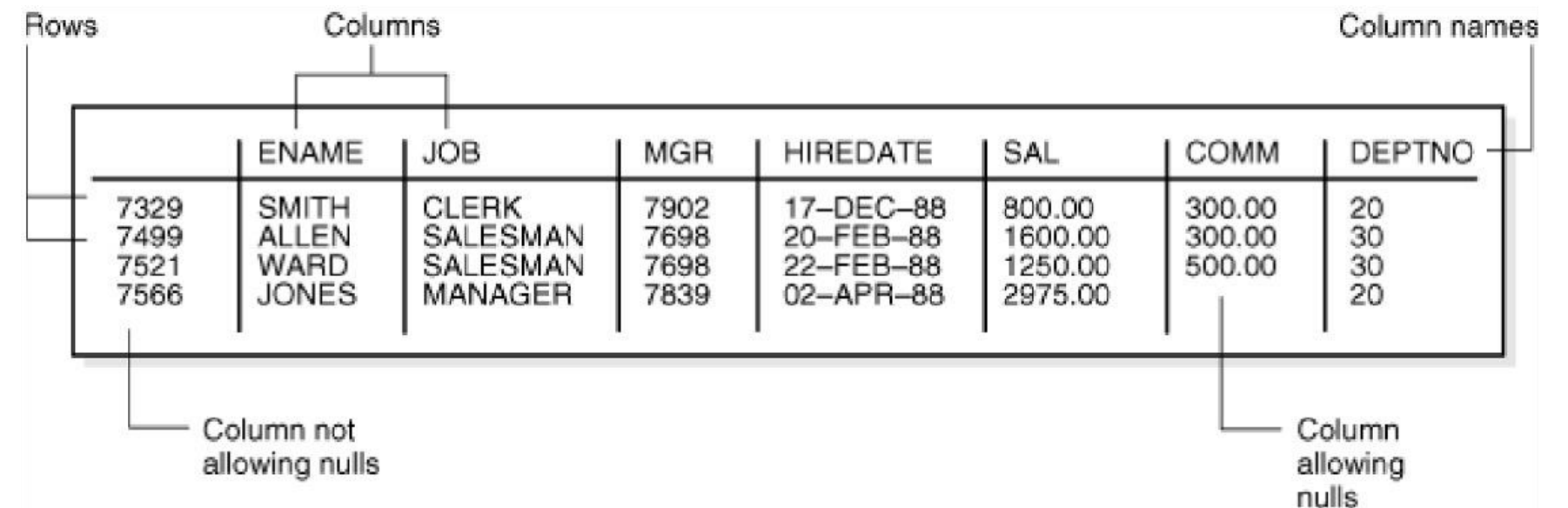
➔ Schema

- A schema is a collection of database objects.
- A schema is owned by a database user and has the same name as that user.
- Schema objects are logical structures created by users.
- Objects such as tables or indexes hold data, or can consist of a definition only, such as a view or synonym.

Oracle Enterprise 19c

➔ Objects

- a. Tables
- b. Views
- c. Sequences
- d. Indexes
- e. Synonyms
- f. Procedures
- g. Functions
- (...)



The diagram illustrates a table structure with the following columns: ENAME, JOB, MGR, HIREDATE, SAL, COMM, and DEPTNO. The table contains four rows of data. Annotations include: 'Rows' pointing to the data rows, 'Columns' pointing to the column headers, 'Column names' pointing to the header row, 'Column not allowing nulls' pointing to the ENAME column, and 'Column allowing nulls' pointing to the COMM column.

	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7329	SMITH	CLERK	7902	17-DEC-88	800.00	300.00	20
7499	ALLEN	SALESMAN	7698	20-FEB-88	1600.00	300.00	30
7521	WARD	SALESMAN	7698	22-FEB-88	1250.00	500.00	30
7566	JONES	MANAGER	7839	02-APR-88	2975.00		20

Oracle Enterprise 19c

➔ Physical Model

Table creation exemple:

```
CREATE TABLE editor
(
    "id_editor" NUMBER(3, 0) NOT NULL ENABLE,
    "name" VARCHAR2(200 byte) NOT NULL ENABLE,
    CONSTRAINT "EDITOR_PK" PRIMARY KEY ("id_editor")
);
```

```
INSERT INTO editora (id_editor, name) VALUES (1, 'Penguin Books');
```


SQL in Oracle

➔ Overview

SELECT	Data retrieval
INSERT UPDATE DELETE MERGE	Data manipulation language (DML)
CREATE ALTER DROP RENAME TRUNCATE	Data definition language (DDL)
COMMIT ROLLBACK SAVEPOINT	Transaction control
GRANT REVOKE	Data control language (DCL)

SQL in Oracle

➔ Overview

- **Views:**

- Virtual table based on a SQL query.
- Simplifies access to complex queries.
- Provides security and reusable logic.

- **Sequences:**

- Generates unique numeric values.
- Commonly used for auto-incrementing IDs.

- **Triggers:**

- Executes automatically on INSERT, UPDATE, or DELETE.
- Enforces rules and audits data changes.

- **Procedures:**

- Stored PL/SQL programs for specific tasks.
- Encapsulates logic and improves reusability.

SQL in Oracle

➔ Overview

```
CREATE VIEW view_name AS
  SELECT columns
  FROM tables
  [WHERE conditions];
```

```
CREATE SEQUENCE sequence_name
  MINVALUE value
  MAXVALUE value
  START WITH value
  INCREMENT BY value
  CACHE value;
```

```
CREATE [OR REPLACE] PROCEDURE procedure_name
  [(parameter_name [IN | OUT | IN OUT] type [, ...])]
  {IS | AS}
  BEGIN
    < procedure_body >
  END procedure_name;
```

```
CREATE [OR REPLACE ] TRIGGER trigger_name
  {BEFORE | AFTER | INSTEAD OF }
  {INSERT [OR] | UPDATE [OR] | DELETE}
  [OF col_name]
  ON table_name
  [REFERENCING OLD AS o NEW AS n]
  [FOR EACH ROW]
  WHEN (condition)
  DECLARE
    Declaration-statements
  BEGIN
    Executable-statements
  EXCEPTION
    Exception-handling-statements
  END;
```

Oracle vs MySQL

Feature	MySQL	Oracle Database
Scalability	Scales well for small to medium workloads. Enterprise edition available for large-scale applications.	Designed for large-scale, mission-critical applications requiring high scalability.
Performance	Good performance for moderate workloads and web applications.	Optimized for large datasets, high transaction volumes, and complex operations.
Data Models	Primarily relational, with support for JSON.	Supports relational, JSON, XML, and other multi-model data types.
Functionality	Basic functions – CRUD (Create, Read, Update, Delete).	Advanced functions – Real Application Clusters (RAC), parallel query execution.
Security	Supports SSL, authorization, and authentication.	Advanced security features, including encryption, data masking, and fine-grained access control.

Oracle vs MySQL

Feature	MySQL	Oracle Database
Storage Engines	Offers different storage engines such as InnoDB and MyISAM for flexibility.	Single, highly optimized storage engine with features like compression and partitioning.
Backup and Recovery	Provides various backup options, including full, partial, and incremental backups.	Offers advanced backup and disaster recovery options like Oracle RMAN (Recovery Manager).
Transaction Support	ACID compliance with the InnoDB engine.	Full ACID compliance, advanced transaction management, and concurrency control.
Cloud Integration	Can be deployed in cloud environments, supported by all major cloud providers.	Fully integrated with Oracle Cloud Infrastructure (OCI), optimized for hybrid and multi-cloud deployments.
Support and Documentation	Strong community support, with optional paid support.	Comprehensive 24/7 enterprise support, extensive documentation, and access to Oracle expertise.

MySQL is an open-source database known for its simplicity, ease of use, and cost-effectiveness. On the other hand, Oracle is an enterprise-level solution offering high scalability and security.

SQL in Oracle

➔ Differences in SQL between MySQL and Oracle

- Record Limitation Syntax:
 - **MySQL:** `LIMIT 10`
 - **Oracle:** Use the `ROWNUM` or `FETCH FIRST N ROWS ONLY` commands.
- Sequence Management:
 - **MySQL:** `AUTO_INCREMENT` em tabelas.
 - **Oracle:** Utiliza objetos `SEQUENCE`

```
CREATE SEQUENCE seq_example START WITH 1 INCREMENT BY 1;
```

SQL in Oracle

➔ PL/SQL

Oracle's unique procedural language for data manipulation and database-side logic.

Basic example:

```
BEGIN
    FOR i IN 1..10 LOOP
        INSERT INTO tabela (coluna) VALUES (i);
    END LOOP;
END;
```


Oracle NoSQL Database

Oracle NoSQL Database is a distributed, highly scalable database designed for unstructured and semi-structured data.

Key Features:

- Supports key-value and document data models.
- High availability and fault tolerance.
- Designed for real-time, large-scale applications.
- Comparison with Oracle Database:
 - Oracle Database: Relational, structured data.
 - Oracle NoSQL: Unstructured/semi-structured, schema-less data.

NoSQL Databases

T2 – Oracle Overview

Teacher: José Machado / Cristiana Neto
Email: jmac@di.uminho.pt /
cristiana.neto@algoritmi.uminho.pt

